# THE NIGERIA ELECTRICITY RESET

## 2026–2031

A 16-Pillar, 6-Phase Execution Blueprint for Solving Nigeria's Power Crisis

*Office of the Citizen*

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*Policy Document — Dual-Layer: Public Doctrine + Technical Annexes*

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# 1. Executive Summary

Nigeria does not have a power generation problem. It has a visibility problem, a cashflow problem, a trust problem, and a governance problem sitting on top of ageing infrastructure. Every reform attempt fails because it treats electricity as engineering when in reality it is economics and behaviour.

This document proposes a 16-pillar, 6-phase electricity reset designed for the actual Nigerian operating environment. It starts not with announcements or megawatts but with control: making the system visible, measurable, and accountable before attempting to fix it. It is authored jointly by Undercurrent and the Office of the Citizen, drawing on NERC quarterly data, World Bank project documentation, the Electricity Act 2023, and sector performance filings through Q2 2025.

The total investment required for a practical five-year programme is approximately $37–49 billion, plus a ₦2.8 trillion legacy debt settlement. This is not the '$100 billion' expansion scenario referenced by the Federal Ministry of Power. It is the highest-return tranche that can realistically be financed, executed, and sustained.

| | |
|---|---|
| **Doctrine** | Visibility → Fairness → Enforcement → Price Truth → Market Discipline |
| **Architecture** | Federal backbone + State execution markets + Industrial power enclaves |
| **Legal foundation** | Electricity Act 2023, Constitution (as amended), NERC Orders and Regulations, PIA 2021 |
| **Core principle** | You cannot fix what you cannot see. In Nigeria, too many actors benefit from the system remaining opaque. |
| **Authorship** | Undercurrent in conjunction with the Office of the Citizen. March 2026. |

## Document Structure

This document operates on two layers. The main paper (Sections 1–19) presents the reform doctrine, phased strategy, institutional architecture, resistance analysis, data platform specification, and workforce programme in language accessible to policymakers, investors, and informed citizens. The technical annexes (A–F) contain implementation-grade detail: matrices specifying who does what under what authority, financing instruments, data system design, workforce deployment plans, DisCo transition protocols, and security frameworks.

New in this edition relative to the first public draft: a Resistance Map (Section 9), a full National Electricity Data Platform architecture (Section 10), a National Electricity Workforce Programme (Section 11), an Irreversibility Architecture (Section 14), and deepened investor

logic in the Financing section (Section 6). Enforcement pathways have been expanded from legal citation to operational capacity across Sections 8 and 9.

# 2. Problem Diagnosis

## 2.1 The Numbers That Matter

As of the most recent verified data (NERC Quarterly Reports Q1–Q2 2025; NERC Annual Report 2024; World Bank DISREP documentation), Nigeria's electricity sector presents the following performance profile:

| Indicator | Value | Source / Period |
|---|---|---|
| Installed generation capacity | ~13,000 MW | FMP / NERC 2024 |
| Average available generation | ~5,396 MW | NERC Q2 2025 |
| Average hourly generation | ~4,501 MWh/h | NERC Q2 2025 |
| Active registered customers | 11.82 million | NERC Q2 2025 |
| Metered customers | 6.42 million (54.33%) | NERC Q2 2025 |
| End-user metering rate (Q4 2024) | 46.57% | NERC Q4 2024 |
| Collection efficiency | 76.07% (Q2); 74.39% (Q1) | NERC 2025 |
| ATC&C losses | 37.92% (Q2); 39.61% (Q1) | NERC 2025 |
| GenCo debt (Feb 2026) | ₦6.8 trillion | FMP / Industry |
| Monthly debt growth | ~₦200 billion | Estimate |
| Grid collapses (2024) | ~12 | NERC 2024 |
| Grid collapses (Jan 2026) | 3 | TCN / media |
| Grid frequency range | 49.46–50.69 Hz (outside safe range) | NERC 2024 |
| Per capita consumption | ~144 kWh/year | World Bank |
| Annual economic cost of outages | ~$29 billion | World Bank est. |
| Generator economy (captive) | ~14 GW, $10bn+ annually | Industry est. |
| Firms leaving grid (2025) | 20+ firms, adding 1,045 MW off-grid | Industry / media |
| DisCo Q1 2025 revenue | ₦553.63bn collected / ₦744.27bn billed | NERC Q1 2025 |

*The gap between installed capacity (~13,000 MW) and average dispatch (~4,500 MW) demonstrates that the crisis is not primarily about building more plants. It is about making the existing system function.*

## 2.2 The Five Failure Layers

Everything wrong with Nigeria's electricity sector falls into five interdependent failure layers. Each layer makes the others worse. Addressing one without the others produces temporary improvement that collapses.

1. Visibility failure. Energy flows are not fully tracked at feeder, transformer, or customer level. Losses hide inside aggregates. Billing is disputed because data is weak. The SCADA/EMS system for grid control remains approximately 70% complete. The end-user metering rate stood at only 46.57% as of Q4 2024, meaning more than half of all customers are on estimated billing.

2. Revenue failure. Collection efficiency dropped to 74.39% in Q1 2025. NERC observed an inverse relationship: when DisCos take more energy, they allocate it to areas with historically lower collection efficiency, meaning expansion actively worsens revenue recovery. Government MDAs are among the worst defaulters.

3. Cost and pricing distortion. Government policy froze end-user tariffs at July 2024 rates even as NERC's MYTO methodology calculates increases for inflation, FX, gas price, and available generation. DisCos collected about ₦570 billion in Q3 2025, largely driven by prior tariff increases rather than improved service.

4. Physical and infrastructure constraints. Distribution networks are weak. The World Bank transmission project had installed 6,330 MVA against a 9,050 MVA target by mid-2025. Grid frequency oscillates between 49.46 Hz and 50.69 Hz, outside the prescribed 49.75–50.25 Hz safe range.

5. Governance and enforcement failure. Nigeria scored 0.897 on the AfDB Regulatory Governance Index 2024 but only 0.642 on regulatory outcomes (service delivery: 0.512). NERC issued 397 regulatory instruments in 2024 but enforcement remains selective. DisCo privatisation agreements created undercapitalised operators shielded by contract.

## 2.3 Recent Developments

GAMCO (March 2026). The Federal Executive Council approved the Grid Asset Management Company Limited to recover stranded generation capacity starting with the Benin-Lagos transmission corridor, aiming to unlock approximately 1,600 MW within 18–24 months. Stakeholders have raised concerns about overlap with NISO (EA 2023, Sections 37–42) and TCN. Legal basis: Executive Order / FEC approval. Not yet anchored in primary legislation.

State electricity markets. At least 16 states have enacted state electricity acts. NERC has transferred regulatory oversight to at least 11 states. Enugu's state regulator issued a tariff order reducing Band A prices, triggering a jurisdictional dispute with NERC over federal vs

state tariff authority. Legal basis: Constitution, Second Schedule, Part II, Item 14; EA 2023, Part XII, Sections 189–199; NERC Transfer Orders.

Proposed DisCo recapitalisation. The draft Electricity Act (Amendment) Bill 2025 would empower NERC to compel core investors in DisCos to inject fresh capital or face share dilution, receivership, or re-privatisation. Over 20 firms left the grid in 2025, adding 1,045 MW off-grid. Legal basis: Current: EA 2023, Sections 73–95. Proposed: EA Amendment Bill 2025. Precedent: BOFIA 2020, Section 12.

NERC Performance Framework. NERC issued the Performance Monitoring Framework Order (July 2024) and Addendum-1 (December 2024). DisCos evaluated on seven KPIs. Failure to offtake 95% of allocated energy in two of three months triggers a 5% downward adjustment to guaranteed admin costs. Legal basis: EA 2023, Section 56(3).

National Integrated Electricity Policy. FEC approved the NIEP on 5 May 2025. Legal basis: EA 2023, Part II, Sections 3–5.

# 3. Reform Doctrine

The reform doctrine rests on one observation: Nigeria's electricity reforms keep failing because they treat the sector as an engineering problem when it is fundamentally a problem of behaviour, incentives, and institutional trust. The correct sequence is not 'build more, hope for the best.' Control the system first, then fix it.

## 3.1 The Doctrine

| | |
|---|---|
| **Phase logic** | Visibility → Fairness → Enforcement → Price Truth → Market Discipline |
| **Meaning** | First make the system measurable. Then make it fair. Then make it hard to cheat. Then make it honest on price. Then let competition do the rest. |
| **Rationale** | Each phase creates the political and institutional preconditions for the next. Enforcement without fairness is perceived as oppression. Price truth without visibility lacks credibility. Market competition without infrastructure is notional. |
| **Non-negotiable** | The sequence is causal, not aspirational. Remove any step and the structure collapses. Every deviation from this sequence in Nigerian electricity history has produced temporary improvement that reversed within two years. |

## 3.2 The Three-Lane Architecture

Nigeria should no longer attempt to serve all electricity needs through a single centralised architecture. The Electricity Act 2023, read with the constitutional amendment placing electricity on the Concurrent Legislative List, provides the legal basis for a three-lane model.

Federal lane. National grid, interstate transmission, system stability, wholesale market rules, large hydro and gas coordination, national energy security, subsidy framework, and the national data platform. Sets standards. Manages backbone infrastructure.

State lane. Intrastate distribution, local embedded generation, mini-grids, urban service improvement, local enforcement, consumer protection, and feeder-level performance. Sixteen states have enacted electricity acts. Three are fully managing their intrastate markets.

Industry lane. Large industrial parks, manufacturers, ports, mines, hospitals, universities, and dense commercial clusters secure dedicated power through captive generation, bilateral contracts, private substations, and special service arrangements. This creates competitive pressure on DisCos.

*Legal basis: Constitution, Second Schedule, Part II, Item 14; EA 2023, Part XII, Sections 189–199 (State Markets); Sections 11–13 (Eligible Customer); Section 62 (Captive Power); NERC Eligible Customer Regulations 2024.*

### 3.3 The Execution Principle

Every element of this plan is designed around one question: who does what, under what authority, funded how, what happens if they fail, and how is failure detected? The implementation matrix in Annex A converts every pillar into an accountable action with a named lead agency, legal basis, funding source, timeline, measurable output, and escalation mechanism. Where escalation is unclear, the reform will be absorbed by the bureaucracy it is trying to change.

# 4. Phased Implementation Strategy

The pillars describe what must be built. The phases describe the sequence. The correct order is causal: each phase creates the conditions for the next. Timelines are indicative; phase transitions are trigger-based (tied to measurable conditions), not calendar-based. A phase that is declared complete before its exit trigger is met simply resets the failure clock.

## Phase 0: Control

*Months 0–6*

Objective: Stop blind operation. Make the system visible before attempting to fix it.

- Establish the Presidential Electricity Reset Council (PERC) by executive order, with a dedicated delivery secretariat, procurement authority, and direct reporting to the Presidency.
- Create the National Electricity Digitisation Office (NEDO) under PERC.
- Launch the National Electricity Data Platform (NEDP): mandatory real-time feeder data from all DisCos, generation data from GenCos, transmission data from TCN. Full architecture in Section 10.
- Begin feeder-level metering on the top 100–200 feeders (highest-revenue and highest-loss).
- Audit real energy flows across the full value chain. Publish findings within 90 days.
- Enforce mandatory data reporting. Non-compliance triggers licence review under EA 2023, Section 76.
- Meter all federal MDAs on prepaid systems with automatic allocation deduction from budget.
- Conduct telecoms readiness assessment for smart metering zones.
- Sign workforce mobilisation contracts for meter installers, SCADA technicians, and feeder monitoring teams. See Section 11.

*Legal basis: PERC: Constitution S.5 (executive powers). NERC enforcement: EA 2023, Sections 73–95. MDA prepaid: Public Finance Management framework.*

**Exit trigger: NEDP operational, top 200 feeders real-time metered, all federal MDAs on prepaid, energy flow audit published.**

## Phase 1: Stop the Bleeding

*Months 6–18*

Objective: Increase cashflow fast. Convert visibility into revenue.

- Aggressive meter rollout in priority zones. Target: 2 million meters in this phase.
- Enforce payment from government, industry, and high-income areas first. Publish MDA payment dashboards.
- Introduce community service compacts in target feeders.
- Launch a 60–90 day amnesty window for illegal connections (one-time only, no extension, no second amnesty).
- Begin public feeder dashboards.
- Activate the National Electricity Ombudsman.
- Initiate DisCo performance assessments under the DisCo Transition Protocol (Annex E).

*Legal basis: Metering: EA 2023, Section 105; MAP Regulations 2021; NMMP Regulations 2021; MAF Orders 2024. Amnesty: proposed NERC Order under EA 2023, Section 96.*

**Exit trigger: Metering coverage exceeds 65% of active customers. Collection efficiency exceeds 80%. Public feeder dashboards operational for the top 200 feeders.**

## Phase 2: Structural Reset

*Months 0–24, parallel*

Objective: Clear legacy debt and secure gas supply, conditioned on Phase 0 completion.

- Create the Power Sector Resolution Vehicle (PSRV), modelled on AMCON. Issue long-dated government-backed notes for verified GenCo arrears (₦2.8tn + $2–3bn working capital).
- Restructure NBET obligations to reflect actual collection capacity.
- Secure gas payment guarantees through a Gas-to-Power Payment Assurance Facility.
- Design subsidy transition architecture: define protected users, build lifeline tariff structure, create direct support channels.
- Establish the Electricity and Energy Infrastructure Division within the Federal High Court.

*Legal basis: PSRV: proposed enabling legislation modelled on AMCON Act 2010. NBET: EA 2023, Section 7. Gas: PIA 2021, Section 104. Court: EA 2023, Section 227; FHC Act, Section 44.*

**Exit trigger: Debt is settled after visibility begins, not before. Major PSRV disbursements conditioned on NEDP operational status.**

## Phase 3: Enforcement

*Months 12–30*

Objective: Make the system hard to cheat. Enforcement begins only after fairness infrastructure is in place.

- Activate the Electricity Division with first prosecutions targeting organised theft syndicates.
- Enforce penalties on underperforming DisCos (Tier 2/3 actions per Annex E).
- No elite immunity. Public prosecutions for major defaulters including MDAs and politically connected debtors.
- Criminal penalties for data falsification by market participants.
- Differentiated enforcement: civil penalties for minor first offences; criminal prosecution for organised sabotage; asset seizure for syndicates.

*Legal basis: EA 2023, Sections 217–226 (offences), Section 227 (jurisdiction). DisCo enforcement: EA 2023, Sections 73–95 and proposed EA Amendment Bill 2025.*

**Exit trigger: Enforcement activates only after metering coverage exceeds 65%, complaint resolution is operational, and billing transition protocol is in place.**

## Phase 4: Infrastructure and Delivery

*Months 12–48*

Objective: Improve actual supply, but only where power can be delivered and paid for.

- Distribution: top 200 feeder rehabilitation ($10–12bn).
- Transmission: complete SCADA/EMS, reinforce corridors around major load centres ($6–8bn).
- Gas: pipeline repairs, compression, redundancy ($5–7bn).
- Generation: add 6–8 GW with signed gas, offtake, and evacuation contracts ($9–12bn). No plant without all three signed.
- Designate Electricity Special Operations Zones (ESOZs) in Lagos, Kano, Aba/Onitsha, Port Harcourt, Abuja.
- Accelerate state electricity market development.

*Legal basis: EA 2023, Parts IV–VII. ESOZs: PERC executive instrument. State markets: EA 2023, Part XII.*

**Exit trigger: ATC&C losses below 25% on priority feeders. Service hours exceed 18 hrs/day in ESOZs.**

## Phase 5: Price Truth

*Months 24–48*

Objective: Remove broad subsidy. Trigger-based, not calendar-based.

- Preconditions: 90%+ metered, 100% feeder metering, MDAs prepaid, collection >85%, service quality audited, complaint resolution functional.
- Major subsidy rollback for metered users with measurable service.
- Full cost-reflective pricing for commercial and industrial users.
- Activate the Power Consumer Assistance Fund (EA 2023, Section 159).

*Legal basis: EA 2023, Section 136 (tariff methodology), Section 159 (PCAF). Proposed: NERC Subsidy Exit Regulations.*

**Exit trigger: Subsidy withdrawal does not proceed until all preconditions are verified by independent audit.**

## Phase 6: Market Evolution

*Months 36–60*

Objective: Long-term stability through competition and decentralisation.

- Fully activate state electricity markets in all qualifying states.
- Transition from single-buyer model (NBET) to bilateral contracts (EA 2023, Sections 6–13).
- Allow embedded generation and captive power to compete through Eligible Customer framework.
- Reduce federal market dominance gradually. Federal lane becomes backbone and standard-setter; states execute; industry self-secures.

*Legal basis: EA 2023, Part III (Competitive Market), Part XII (State Markets); NERC Eligible Customer Regulations 2024.*

# 5. Institutional Architecture

Every reform pillar requires an institutional home. This section specifies the new institutions required, clarifies the mandates of existing agencies, and lists the legislative instruments needed.

## 5.1 New Institutions Required

Presidential Electricity Reset Council (PERC). Apex coordination body. Chaired weekly by VP or Chief of Staff. Membership includes the Ministry of Power, NERC, TCN, NISO, NBET, Ministry of Finance, CBN, gas suppliers, state electricity commissioners (rotating), manufacturers' association, and security agencies. PERC coordinates delivery; NERC regulates. The relationship is parallel, not hierarchical. PERC's secretariat holds procurement authority and a ring-fenced delivery budget. Established by executive order under Constitution S.5.

National Electricity Digitisation Office (NEDO). Reports to PERC. Owns the design, procurement, deployment, and operation of the NEDP, feeder metering systems, and smart metering backbone. Fixed-term delivery unit (5-year mandate) with ring-fenced procurement authority and independent audit obligations. Transitions data platform ownership to NISO or an independent Data Authority at end of mandate. Established by PERC executive instrument.

Power Sector Resolution Vehicle (PSRV). Special purpose entity for legacy debt settlement. Issues long-dated government-backed notes. Modelled on AMCON (AMCON Act 2010). Board includes the Ministry of Finance, CBN, NERC, and independent directors. Sunset clause: 7-year wind-down. Requires enabling legislation.

National Electricity Ombudsman (NEO). Independent consumer protection body reporting to the National Assembly, not NERC or the Ministry. Powers: order refunds, impose penalties on DisCos, publish quarterly performance reports. Appointed by the President on recommendation of the National Assembly. Fixed 5-year term. Funded by market participant levy, not Ministry or NERC budgets. Requires the proposed National Electricity Ombudsman Act.

Electricity and Energy Infrastructure Division. Designated division within the Federal High Court. Strict procedural timelines (14 days to first hearing). Designated judges and prosecutors with forensic utility evidence training. Judgment window: 90 days. Heavily restricted interlocutory abuse. Requires Practice Direction from the Chief Judge (FHC Act, Section 44) or EA 2023 amendment.

Gas-to-Power Payment Assurance Facility. Ring-fenced facility guaranteeing payment to gas suppliers. Funded through tariff recovery, sovereign guarantee, and DFI credit enhancement. Established by regulation under EA 2023 and PIA 2021.

National Electricity Skills Certification Board (NESCB). Proposed new body housed within NEMSA. Sets national standards for training, certification, and accreditation across all technical trades in the electricity sector. Required to validate training programmes at polytechnics, TVET institutions, and DisCo in-house academies. See Section 11.

## 5.2 Legislative Requirements

The following new legislative or regulatory instruments are required. Most can be fast-tracked as executive instruments or NERC regulatory orders. Only three require National Assembly passage.

| Instrument | Type | Purpose | Status |
|---|---|---|---|
| Electricity Act (Amendment) Bill 2025 | Primary legislation | Compulsory DisCo recapitalisation, share dilution, receivership powers for NERC | Draft — before National Assembly |
| National Electricity Ombudsman Act | Primary legislation | Establish NEO as independent body with refund/penalty powers | Proposed — new |
| PSRV Enabling Act | Primary legislation or executive order | Establish PSRV for legacy debt settlement | Proposed — new |
| NERC Subsidy Exit Regulations | Regulatory instrument | Define measurable preconditions and terminal conditions for subsidy withdrawal | Proposed — new |
| NERC Data Governance Regulations | Regulatory instrument | Mandatory data submission, formats, audit, cybersecurity, falsification penalties | Proposed — new |
| Federal-State Tariff Interface Rules | Regulatory instrument | Clarify how state tariff orders interact with wholesale costs | Proposed — new |
| FHC Practice Direction | Judicial instrument | Establish Electricity Division with procedural timelines | Proposed — new |
| PERC Executive Order | Executive instrument | Establish Presidential Electricity Reset Council | Proposed — new |
| MDA Prepaid Metering Directive | Executive/budget directive | Mandate prepaid for all federal MDAs | Proposed — new |
| National Electricity Skills Certification Regulations | Regulatory instrument | Mandate certification standards and contractor training quotas | Proposed — new |

# 6. Financing Architecture

A $37–49 billion programme cannot be funded from budgetary allocation alone. The financing architecture is designed around a principle: public money de-risks the system; concessional capital bridges the gap; private capital flows once tariff reform and visibility make returns predictable.

## 6.1 Total Cost Envelope

| Component | Estimated Cost | Financing Category |
|---|---|---|
| Legacy debt and working-capital reset | ₦2.8tn + $2–3bn | Sovereign / Concessional |
| Universal metering | $1.4–1.8bn | Blended / Vendor |
| Distribution rehab | $10–12bn | Concessional / Blended |
| Transmission and SCADA | $6–8bn | Sovereign / Concessional |
| Gas-to-power infrastructure | $5–7bn | Blended / IOC equity |
| New generation (6–8 GW) | $9–12bn | Commercial / Private |
| Off-grid and state markets | $3–5bn | Concessional / Mini-grid IPPs |
| Digitisation and monitoring | $2–4bn | Concessional / Tech partners |
| Workforce mobilisation | $0.2–0.4bn | Sovereign / Embedded |
| TOTAL | ~$37–49bn + ₦2.8tn | |

## 6.2 Bankability Classification

| | |
|---|---|
| **Bankable now** | New generation (signed gas/offtake), embedded/captive power, mini-grids in commercially viable corridors, vendor-financed metering. |
| **Bankable after tariff reform** | Distribution rehabilitation, DisCo recapitalisation equity, large-scale feeder modernisation. |
| **Public-good only** | Transmission backbone, SCADA/EMS, grid security, national data platform, legacy debt settlement. |
| **Subsidy-supported** | Rural off-grid, lifeline tariff coverage, clinic/school solar, state market capacity building. |

## 6.3 Financing Sources

Sovereign bonds / promissory notes. For legacy debt via PSRV. Long-dated (7–10 year) naira-denominated. Precedent: FGN promissory notes; AMCON bonds.

DFI concessional lending. World Bank (~$2bn invested in last five years including DARES at $750m), AfDB, IFC. Target: transmission, distribution, metering, SCADA, off-grid.

Blended finance vehicles. SPVs with first-loss tranches from DFIs, senior debt from commercial lenders. Target: distribution, gas, metering at scale.

Private equity / project finance. For bankable generation. IPP model with signed PPA, gas supply agreement, and grid connection. Precedent: Azura-Edo IPP (461 MW).

Carbon credit monetisation. For renewable and distributed energy components under Article 6 of the Paris Agreement or voluntary markets.

Climate finance. Green Climate Fund, Climate Investment Funds, bilateral climate finance. Requires credible decarbonisation pathway.

Performance-linked guarantees. Partial risk guarantees from IDA/MIGA covering regulatory and payment risk. Precedent: World Bank PSRO programme.

## 6.4 Why Private Capital Would Actually Come Under Nigerian Conditions

The standard development-finance answer to this question is 'de-risk and they will come.' That is insufficient for Nigeria. The following analysis addresses the specific risks that have historically kept private capital away, and how this plan's sequencing addresses each.

### Tariff Risk

Nigeria's history of political interference in tariff setting — most recently the freeze at July 2024 rates — is the primary deterrent to long-term private generation investment. The plan addresses this through three mechanisms. First, trigger-based subsidy exit regulations (proposed NERC instrument) remove the political discretion to freeze tariffs indefinitely: once metering, collection, and service conditions are met, NERC's MYTO methodology automatically drives tariff adjustments. Second, the Performance Monitoring Framework's evidence base makes it harder to claim consumer protection justifications for freezes: public dashboards show actual service hours and billing accuracy. Third, Eligible Customer Regulations 2024 create a parallel market at cost-reflective pricing for large consumers, proving commercial tariffs are sustainable and reducing political incentive to keep the general tariff distorted.

### FX Risk

Nigeria's generator economy runs overwhelmingly on dollar-denominated fuel costs while revenue is naira-denominated. For new generation IPPs, FX risk is managed through: (1) gas-supply agreements with domestic producers denominated in naira, reducing the dollar-cost exposure of fuel; (2) PPA structuring with indexed escalation clauses; (3) partial risk guarantees from MIGA or equivalent covering transfer and convertibility risk; (4) where

possible, eligible customer contracts with industrial offtakers who have dollar revenue streams, which removes government as the only counterparty.

### Payment Risk

The standard Nigerian IPP complaint is that NBET does not pay. The Gas-to-Power Payment Assurance Facility addresses this for gas suppliers. For generation investors, the PSRV restructures the existing payment backlog, and NBET's obligations are restructured to reflect actual collection capacity. The Payment Waterfall (see Annex B) ring-fences generation payments at the top of the collection cascade, ahead of DisCo operating costs. World Bank PSRO precedent provides partial risk coverage for payment default.

### What Becomes Bankable in Each Phase

Phase 0: Vendor-financed metering (MAP model) is bankable immediately because DisCo license conditions compel meter procurement and repayment is structured through tariff recovery. Eligible customer generation contracts for large industrial offtakers are bankable now.

Phase 1: Mini-grids in commercially viable corridors become bankable as community service compacts and amnesty regularisation clarify the customer base and expected revenue. Off-grid solar for anchor customers (hospitals, factories) achieves bankability with state government guarantees.

Phase 2–3: Once the PSRV clears the legacy debt and feeder dashboards make collection efficiency verifiable, distribution rehabilitation becomes bankable to blended finance vehicles. DisCo recapitalisation equity becomes investable for restructuring-oriented funds once the EA Amendment Bill gives NERC the powers to enforce outcomes.

Phase 4–5: Large-scale new generation (6–8 GW) becomes fully bankable once tariff reform is triggered by metering preconditions, removing the political discretion that has historically killed Nigerian IPP financings at the signing stage.

# 7. Market Restructuring: Federal / State / Industry

The three-lane architecture requires specific rules at each level. This section specifies the scope, regulator, tariff authority, and data obligations for each lane, and addresses the critical federal-state tariff interface.

## 7.1 Federal Lane

| | |
|---|---|
| **Scope** | National grid, interstate transmission, system stability, wholesale market, NISO, large hydro, gas coordination, NBET/settlement, national data platform, subsidy framework. |
| **Regulator** | NERC (EA 2023, Sections 31–72). |
| **Market operator** | NISO (EA 2023, Sections 37–42). |
| **Tariff** | NERC sets wholesale tariffs via MYTO methodology (EA 2023, Section 136). |
| **Market evolution** | Single-buyer (NBET) → bilateral contracts → full competitive market (EA 2023, Sections 6–13). |

## 7.2 State Lane

| | |
|---|---|
| **Scope** | Intrastate distribution, local embedded generation, mini-grids, off-grid, local enforcement, consumer protection. |
| **Regulator** | State Electricity Regulatory Commission (SERC), with oversight transferred from NERC under EA 2023, Part XII. |
| **Tariff** | SERCs set retail tariffs. Critical issue: interface with federal wholesale tariff requires proposed Federal-State Tariff Interface Rules. |
| **Data obligation** | SERCs must integrate data into the NEDP. National visibility is non-negotiable even when regulatory authority is decentralised. |

## 7.3 Industry Lane

| | |
|---|---|
| **Scope** | Large consumers with dedicated power (industrial parks, manufacturers, ports, mines, hospitals, universities). |
| **Legal framework** | EA 2023, Section 62 (captive power); NERC Eligible Customer Regulations 2024 (five customer classes); Sections 11–13. |
| **Competitive pressure** | Over 20 firms left the grid in 2025, adding 1,045 MW off-grid. This is market discipline in action. The plan channels this rather than ignoring it. |

## 7.4 The Federal-State Tariff Interface

The most politically significant unresolved issue is tariff jurisdiction. The Enugu case — where the state regulator issued a Band A order below the NERC-approved rate — will recur in multiple states. This plan proposes three binding principles for resolution:

| | |
|---|---|
| **Principle 1** | States cannot set retail tariffs below wholesale generation cost plus transmission charges. States can design their own retail structures above or at cost-reflective levels. |
| **Principle 2** | Any state subsidy must be funded by the state, not absorbed by GenCos, NBET, or TCN. |
| **Principle 3** | Data transparency is the enforcement mechanism. Wholesale payment compliance is verified through the NEDP. States that cannot demonstrate compliance lose NERC transfer authority. |

# 8. Enforcement and Legal Framework

Enforcement is the hinge of the entire reform. Without it, visibility becomes decoration and price truth becomes exploitation. But enforcement must come after fairness is established. Enforcing payment discipline on a system citizens have legitimate reason to distrust is not reform; it is extraction.

## 8.1 Existing Legal Framework

| Offence / Power | Legal Basis |
|---|---|
| Electricity theft | EA 2023, Section 217: fine and/or minimum 3 years imprisonment. |
| Damage to installations | EA 2023, Section 218. |
| Meter tampering | EA 2023, Sections 219–222. |
| Illegal connections | EA 2023, Sections 223–224. |
| Jurisdiction | EA 2023, Section 227: Federal High Court. |
| NERC enforcement | EA 2023, Sections 73–95: rectification directives, financial penalties, licence revocation (Section 76). |
| Pipeline protection | PIA 2021, Section 215; Miscellaneous Offences Act. |

## 8.2 Proposed Enhancements

### Electricity and Energy Infrastructure Division

Jurisdiction covers electricity theft, meter tampering, illegal connections, transformer vandalism, gas pipeline sabotage, non-remittance fraud, data falsification, infrastructure sabotage, and staff collusion. Procedural rules: filing to first hearing within 14 days; interlocutory abuse heavily restricted; judgment window of 90 days; designated prosecutors with forensic utility evidence training.

### Differentiated Enforcement Regime

| Category | Response |
|---|---|
| Minor first offence | Civil penalty, regularisation, mandatory metering. No criminal record. |
| Repeat theft | Heavy financial penalty, disconnection, civil recovery. |
| Organised sabotage | Criminal prosecution, minimum 3 years (EA 2023, S.217). Aggravated sentencing for syndicate leaders. |
| Syndicates / contractors | Asset seizure. Criminal prosecution of all participants. |

| | |
|---|---|
| Staff collusion | Internal prosecution plus criminal referral. Disqualification from sector employment. |
| Data falsification | Proposed new offence: equivalent to financial fraud (Cybercrimes Act 2015). Criminal prosecution regardless of position. |
| MDA / government default | Public naming. Automatic budget deduction. PERC escalation to Presidency. No discretionary waiver. |

## Investigation and Evidence Pathway

Enforcement without an investigation pathway is announcement, not deterrence. This plan specifies the operational sequence from detection to prosecution.

Detection. Anomaly detection on the NEDP flags statistical deviations at feeder level: unexplained energy losses, billing-to-collection gaps, meter data discontinuities, and submission failures. Tamper alerts from smart meters route directly to the enforcement queue. Community compact networks provide ground-level intelligence on bypass networks. NEO complaint data identifies DisCos with systemic billing abuse patterns.

Investigation. NERC's enforcement directorate conducts primary investigations for DisCo-level violations. The Economic and Financial Crimes Commission has concurrent jurisdiction for non-remittance fraud above a threshold of ₦100 million. NSCDC holds primary jurisdiction for physical infrastructure sabotage. Inter-agency protocol is specified in a Joint Investigation Framework established at Phase 0.

Evidence gathering. Forensic meter data (tamper logs, load profile, disconnection history) is extracted from the NEDP chain of custody. On-site evidence collection by NEMSA-certified inspectors. Chain of custody is documented digitally from extraction through Electricity Division filing. Falsified data cannot be substituted: the NEDP's immutable logging architecture (see Section 10) ensures original records are preserved for prosecution regardless of subsequent DisCo data submission.

Case prioritisation. Three categories: Category A — Organised sabotage and theft syndicates, data falsification by DisCo officers, non-remittance fraud above ₦100m. Target: prosecution within 90 days of filing. Category B — Repeat individual theft, meter tampering with commercial intent, illegal connection after amnesty window. Target: resolution within 180 days. Category C — Minor first offences, estimated billing disputes with compliance. Target: civil resolution within 30 days through NEO.

Anti-selective enforcement. Each quarter, NERC publishes an enforcement action register covering all investigations opened, categories, outcomes, and — critically — which cases were escalated, stalled, or dropped, and the reason. Any case involving an MDA, a political figure, a military installation, or a government contractor that is stalled for more than 60 days

without documented justification triggers automatic referral to the Attorney General. Public dashboards show enforcement by DisCo zone, enabling comparative scrutiny.

### Sequencing Principle

Enforcement (Phase 3) activates only after fairness infrastructure is verified: metering coverage exceeding 65%, billing transition protocol operational, Ombudsman functional, complaint resolution system measured and published. This sequencing is the political backbone of the entire doctrine. Citizens who have been metered, whose bills are accurate, and who have a functioning complaint channel have no legitimate grievance when enforcement arrives.

# 9. Resistance Map: Who Loses and How They Fight Back

Every reform in this plan threatens a specific interest. This is not an abstract observation. It is an operational input. Identifying the resistance in advance determines what counter-design is required. The following map covers the primary resistance actors, their likely tactics, and the structural countermeasures built into this plan.

## 9.1 Distribution Companies (DisCos)

What they lose: Opacity protection. The current system allows DisCos to hide ATC&C losses inside aggregated numbers, dispute bills without resolution, and resist recapitalisation by claiming the problem is tariff distortion rather than management failure. Feeder-level public dashboards, the Performance Monitoring Framework, and the DisCo Transition Protocol's quantitative thresholds eliminate these defences.

How they resist:

- Litigation. DisCos will challenge NERC enforcement actions, recapitalisation orders, and PMF penalties in court. Interlocutory injunctions have historically stalled regulatory action for years.
- Data non-submission. Selective or falsified data submission to the NEDP, exploiting technical complexity to dispute anomaly detection findings.
- Metering delay. Slow-walking meter installation by citing procurement bottlenecks, vendor failures, or customer refusal.
- Narrative capture. Lobbying media and legislators with the argument that regulation is strangling private investment; framing recapitalisation as expropriation.

Countermeasures:

- The Electricity Division's 14-day filing to first-hearing rule and restricted interlocutory scope limits delay-by-litigation.
- Licence conditions make NEDP data submission a non-discretionary obligation. Non-submission triggers automatic licence review (EA 2023, S.76), not a negotiation.
- The EA Amendment Bill removes the legal ambiguity DisCos exploit by creating an explicit recapitalisation power with clear procedural safeguards. Shareholder challenges are heard by the Electricity Division, but challenges do not stay the process.
- PMF public reporting and feeder dashboards create independent verification that makes DisCo narrative capture expensive. Media, civil society, and the Ombudsman all have access to the same underlying data.

## 9.2 Government Ministries, Departments, and Agencies (MDAs)

What they lose: The ability to accumulate electricity debt without consequence. MDAs are currently among the largest electricity defaulters in the country. Prepaid metering with automatic budget deduction eliminates this.

How they resist:

- Bureaucratic delay. Slow-walking prepaid meter installation through procurement objections, ministerial non-response, and referral loops.
- Political override. Ministerial intervention to exempt specific agencies from prepaid requirements, citing security, health, or operational sensitivity.
- Administrative obstruction. Refusing to share billing accounts with OAGF, disputing meters' accuracy, or claiming technical incompatibility.

Countermeasures:

- MDA prepaid metering is established by executive directive from the Presidency, not NERC or the Ministry of Power. The authority sits above the resistance.
- Automatic budget deduction through the OAGF removes discretion from the payment decision. The meter deducts; the payment happens.
- PERC weekly reporting makes MDA payment performance visible to the Chief of Staff. Non-compliance is a governance failure visible at apex level.
- Public MDA payment dashboards expose the largest defaulters by name. Political cost of resistance increases with visibility.

## 9.3 Politically Protected Defaulters

What they lose: The ability to avoid bills by political proximity. Military installations, government guest houses, political estates, and party-linked businesses have historically received electricity without payment.

How they resist:

- Direct political intervention. Phone calls from political offices to DisCo management instructing them not to disconnect.
- Security interference. NERC inspectors and DisCo enforcement personnel being warned off by security personnel at sensitive locations.
- Procurement leverage. Politically connected defaulters threatening to use their influence over DisCo licence renewals, contract awards, or regulatory appointments.

Countermeasures:

- Public debt naming. The NEDP's MDA payment dashboard includes all categorised accounts — military, government, political — without exemption. Naming is the first enforcement step; it costs the reform nothing and is politically defensible.
- PERC's membership includes security agencies. Interference with enforcement operations is escalated through the Council's security agency representatives to command level.
- The enforcement sequencing principle means this layer is addressed after visibility establishes the facts publicly. By the time enforcement reaches political estates, the data is already in public dashboards, making selective enforcement politically costly.

## 9.4 Internal Collusion Networks

What they lose: The revenue streams generated by colluding with bypass networks, falsifying energy readings, and facilitating illegal connections for payment. DisCo staff at feeder and distribution levels are frequently embedded in commercial theft arrangements.

How they resist:

- Meter data manipulation at source, before upload to NEDP.
- Tipping off bypass networks when inspections are scheduled.
- Fabricating customer complaint logs to show false compliance with NEO timelines.
- Passive obstruction: meter installers who 'lose' equipment, SCADA technicians who misreport readings, data officers who submit incomplete files.

Countermeasures:

- The NEDP's immutable audit log (see Section 10) means original meter data cannot be overwritten. Attempts to submit falsified corrections are flagged as anomalies.
- Tamper-alert meters route directly to the enforcement queue, bypassing DisCo management.
- Data falsification is a proposed criminal offence equivalent to financial fraud. Individual staff liability, not just corporate liability, is the deterrent.
- Whistle-blower protection under the proposed NERC Data Governance Regulations creates an internal counter-network.
- Staff collusion cases are handled by the Electricity Division and include disqualification from sector employment for convicted individuals.

## 9.5 Illegal Connection Syndicates

What they lose: Revenue from commercial bypass operations. In dense urban zones, organised networks install bypasses for payment, service political clients, and extract fees from customers who cannot afford formal connections.

How they resist:

- Violence and intimidation against meter installers in their territory.
- Bribing local DisCo staff to delay installation or report false metering.
- Community narrative framing themselves as providing affordable power to poor households that the DisCo ignores.

Countermeasures:

- The 60–90 day amnesty window is designed precisely to separate commercial syndicates from households who informally connected out of necessity. Legalising the household connections removes the syndicate's community cover.
- Post-amnesty, the Electricity Division's jurisdiction includes asset seizure for syndicate operators. Criminal prosecution is not discretionary.
- Community service compacts give organised communities a stake in the formal system, creating a social incentive to report bypass activity in their feeder zone.
- The Power Infrastructure Security Framework (Annex F) includes intelligence coordination between NSCDC, police, and DisCos specifically targeting organised bypass networks.

## 9.6 Contractors and Emergency-Intervention Beneficiaries

What they lose: The recurring procurement flow generated by a system that constantly fails and requires emergency repair. Transformer replacement contractors, emergency diesel suppliers, and meter-procurement middlemen benefit from dysfunction continuing.

How they resist:

- Lobbying against metering frameworks that bypass traditional procurement routes (MAP model reduces DisCo discretion in meter procurement).
- Challenging PERC's procurement authority in court, arguing interference with DisCo operational independence.
- Narrative attacks on digitisation — claiming data platforms are surveillance tools or data security risks — to delay NEDP deployment.

Countermeasures:

- NEDO's ring-fenced procurement authority and competitive tender processes reduce individual contractor leverage.
- The mandatory workforce training requirements in contracts above $10 million create a new contractor class with aligned interests in programme continuity.
- Transparent procurement dashboards — part of the NEDP public layer — expose contract values, timelines, and delivery against specification.

## 9.7 State Governments and Governors

What they lose (selectively): Tariff discretion. Governors who have used electricity as a political tool — promising cheap power as a populist offer — lose the ability to set retail tariffs below cost without funding the subsidy from their own budgets.

How they resist:

- Issuing SERC tariff orders below federal wholesale cost, as Enugu has done, to demonstrate political independence.
- Refusing to integrate SERC data into the NEDP, claiming state regulatory autonomy.
- Lobbying the National Assembly against the Federal-State Tariff Interface Rules.

Countermeasures:

- Principle 3 of the Tariff Interface framework: NERC transfer authority is conditioned on data integration compliance. States that refuse NEDP integration lose the regulatory autonomy transfer they sought.
- Principle 2: state subsidies must be funded by the state. A governor who sets below-cost tariffs and cannot fund the difference faces the political cost of that choice domestically.
- The Forum of Regulators (inaugurated March 2026) provides a coordinating body where NERC and SERCs resolve interface disputes before they become litigation.

# 10. National Electricity Data Platform: Full Architecture

The National Electricity Data Platform (NEDP) is the operational foundation of the entire reform. Without it, metering is a procurement exercise rather than a visibility tool, enforcement has no reliable evidence base, and tariff reform is politically undefendable. The NEDP must be designed to produce data that is reliable enough to carry the weight of enforcement, transparent enough to sustain public trust, and technically robust enough to survive attempted manipulation.

This section specifies the architecture at the level of detail required to commission, procure, and govern the platform. It is not a technology recommendation document; it is a policy specification that any credible technical implementation must satisfy.

## 10.1 Platform Ownership and Governance

| | |
|---|---|
| **Build phase owner** | NEDO (National Electricity Digitisation Office), Months 0–60. |
| **Operational owner** | Transitions to NISO or an independent National Electricity Data Authority at end of NEDO mandate. Transition plan published at Month 36. |
| **Regulator** | NERC. Sets data submission standards, validates compliance, issues penalties for non-submission. |
| **Independent auditor** | Third-party firm, competitively procured, rotated every 3 years. Quarterly integrity audits. Results published. |
| **Cybersecurity oversight** | Dedicated team within NEDO. Annual external penetration test. Findings published in redacted form. |

## 10.2 Three-Layer Architecture

The NEDP operates across three distinct data layers, each with different sources, frequencies, access controls, and public disclosure rules.

### Layer 1: Generation and Transmission

Sources: GenCos, TCN, NISO. Data captured: generation output by plant and unit (MWh per hour), available capacity versus dispatched capacity, frequency and voltage readings at key grid nodes, trip and collapse events with timestamp and cause, gas supply volumes and pressures at generating station metering points, interconnector flows, reserve margins.

Upload frequency: Hourly for operational data. Daily summary by 06:00 the following day. Real-time feed for grid frequency, voltage, and event data routed directly to NISO operations.

Access: NISO and NERC have full real-time access. TCN has read access. Public dashboard shows daily aggregates: generation by source type, availability rate, grid frequency range, collapse events.

### Layer 2: Distribution and Retail

Sources: DisCos, SERCs (where applicable). Data captured: energy received at each injection substation (MWh by feeder), energy billed and energy collected at feeder level, ATC&C loss rate per feeder, billing cycle data by customer class, complaint volumes and resolution rates by DisCo zone, meter coverage rate by feeder, service hours by feeder and band.

Upload frequency: Feeder injection data — hourly for the top 200 priority feeders; daily for all others. Billing and collection data — monthly, filed within 15 days of end of billing cycle. Complaint data — weekly. Non-submission triggers automatic PMF penalty at end of quarter.

Access: NERC and NEDO have full access. State regulators have read access to their own SERCs' feeders. Public dashboard shows per-feeder data: energy received, energy billed, energy collected, hours supplied, loss percentage. Updated daily for priority feeders, weekly for all others.

### Layer 3: Customer and Metering

Sources: DisCos, Meter Asset Providers (MAPs), meter data management system (MDMS). Data captured: meter serial number and location (feeder, transformer, coordinates), meter reading by billing cycle, tamper alerts (date, type, severity), disconnection and reconnection events, load profile where smart meters are installed, customer tariff band assignment, complaint log by account.

Upload frequency: Meter reads — monthly by billing cycle. Tamper alerts — real-time, routed directly to the enforcement queue. Disconnection events — within 24 hours. Load profile data from smart meters — daily.

Access: NERC and NEDO have full access for audit. NEO has access to complaint and billing data for dispute resolution. Customers access their own account data through a public portal. Raw individual customer data is not in the public dashboard — only aggregated feeder statistics.

## 10.3 Data Standards and Interoperability

Open API standard. The NEDP uses RESTful APIs with defined schema for all data submissions. No proprietary data format is accepted. All submission specifications are published and version-controlled. DisCos, GenCos, TCN, and SERCs submit against the published schema; NEDO validates format on receipt.

No proprietary lock-in. The platform is procured with open architecture requirements. No single vendor may hold exclusive rights to any data format, submission protocol, or processing layer. Platform source code is owned by NEDO.

Interoperability across agencies. NISO has read access to Layer 1 through a defined API endpoint. SERCs have read access to their jurisdiction's Layer 2 data through state integration nodes. The NEO has a defined read interface for Layer 3 complaint and billing data. EFCC and the Electricity Division's prosecution team have a secure evidence-export interface for legally specified data extractions with chain-of-custody documentation.

Meter data management system (MDMS). The MDMS sits within NEDO and integrates with DisCo billing systems. All meter reads are processed through the MDMS before billing is generated. This breaks the current practice of DisCos generating bills independently of verified meter data.

## 10.4 Validation, Anomaly Detection, and Data Integrity

Automated validation on submission. Every data submission is validated in real time against four checks: format compliance (schema match); range plausibility (no physically impossible values); temporal consistency (readings cannot decrease over time for cumulative meters; sudden drops flag for review); and cross-layer coherence (energy received at feeder must reconcile with billing data within a tolerance band).

Statistical anomaly detection. Weekly automated scan across all feeder data identifies: unexplained loss spikes (greater than 15 percentage points in any 30-day window); billing-to-collection ratios that diverge sharply from historical baseline; submission patterns that suggest fabrication (too-regular readings, round-number clustering). Anomalies are flagged to NERC's enforcement directorate and the Ombudsman within 48 hours.

Immutable audit log. All submitted data is written to an append-only ledger. Corrections are permitted only through a formal amendment process that preserves the original submission alongside the correction and the reason. No data can be deleted. The audit log is available to the independent third-party auditor and, on legal request, to the Electricity Division's prosecution team.

Data falsification detection. The immutable log means that attempting to falsify historical data requires amending records that already exist in the audit trail. Discrepancies between original submissions and amended submissions are flagged. Patterns of systematic amendment in a single direction trigger an NERC investigation referral. Data falsification is a proposed criminal offence equivalent to financial fraud.

Chain of custody for enforcement. Extractions of data for prosecution purposes are conducted through a documented legal interface: the requesting authority (Electricity Division, EFCC, NSCDC) files a data request; NEDO produces a certified extract with hash

verification; the extract cannot be modified after certification; the chain of custody is recorded in the audit log. This makes NEDP data admissible as digital evidence under the Cybercrimes Act 2015 and the Evidence Act.

## 10.5 Cybersecurity Requirements

Classification. The NEDP is classified as critical national infrastructure. Security requirements are equivalent to those applied to the CBN's financial system infrastructure.

Minimum technical requirements: end-to-end encryption for all data in transit and at rest; role-based access control with multi-factor authentication for all agency users; network segmentation between public dashboard layer and operational data layer; intrusion detection and prevention systems; real-time alerting on anomalous access patterns.

Annual external penetration test. Conducted by an independent firm, competitively procured, rotating every two years. Findings reported to NEDO and NERC. Critical findings must be remediated within 30 days; medium findings within 90 days. Summary findings are published in redacted form.

Incident response. NEDO maintains a documented incident response plan. Any breach affecting operational data triggers notification to NERC within 4 hours and public disclosure within 72 hours. Recovery time objective for operational data: 4 hours. Recovery point objective: 1 hour.

## 10.6 Connectivity and Fallback Mechanisms

Phase 0 telecoms assessment. Before deployment, NEDO conducts a zone-by-zone assessment of available connectivity: mobile data coverage by carrier, fibre availability at substations, VSAT feasibility for remote locations.

Connectivity hierarchy. Priority 1: fibre at substation (lowest cost, highest reliability). Priority 2: 4G/LTE cellular modem with SIM failover. Priority 3: satellite (VSAT or low-earth orbit) for remote and underserved zones.

Mesh network fallback. In zones where no reliable connection is available, store-and-forward mesh networks allow local meter data to be aggregated at a node and uploaded when connectivity is restored. Upload delay tolerance: up to 72 hours for non-priority feeders. Priority feeders must have primary plus backup connectivity.

Manual submission protocol. Where electronic submission fails for more than 72 hours, DisCos submit a paper-equivalent data extract to NERC's regional office. This counts as submission for compliance purposes but triggers an infrastructure review at the next PERC weekly meeting.

## 10.7 Public Dashboard vs Regulator-Only Data

| Dashboard type | Content and access |
|---|---|
| Public feeder dashboard | Energy received, energy billed, energy collected, hours supplied, loss percentage per feeder. Updated daily (priority) / weekly (all). Accessible to any member of the public. Downloadable as open data. |
| MDA payment dashboard | All government ministry, department, and agency accounts: balance outstanding, payment history, last payment date, automatic deduction status. Public. Named by institution. |
| Enforcement action register | All NERC investigations opened, by category, outcome, and status. Cases involving MDAs, government contractors, and political estates flagged. Public. |
| NERC regulator dashboard | Full Layer 1, 2, 3 data. Anomaly flags. Investigation queue. DisCo PMF scores. Not public. |
| NEO case dashboard | All complaints received by DisCo, resolution rate, penalties imposed, refunds ordered. Public. |
| Electricity Division case tracker | Cases filed, hearings scheduled, judgments delivered. Public summary only; case details subject to court rules. |

# 11. National Electricity Workforce Programme

Nigeria's electricity sector cannot be reformed without the people to execute the reform. The metering programme requires certified meter installers. The SCADA completion requires qualified technicians. The data platform requires operators with genuine data skills. The Ombudsman requires trained case officers. The Electricity Division requires prosecutors with forensic utility evidence expertise. None of these people exist in sufficient numbers today.

The National Electricity Workforce Programme (NEWP) is a coordinated, five-year effort to build the technical human capital the sector requires, housed within the National Electricity Skills Certification Board (NESCB) under NEMSA, and delivered through partnerships with existing institutions rather than building a new parallel infrastructure.

## 11.1 Governance Structure

National Electricity Skills Certification Board (NESCB). Proposed new body within NEMSA (EA 2023, Section 82). Mandate: set national competency standards for all technical roles in the electricity sector; accredit training institutions; issue certifications; maintain a national skills register; and validate compliance with contractor training requirements.

NEWP Secretariat. Operational unit within NEDO, reporting to PERC. Coordinates training delivery, manages institutional partnerships, tracks deployment targets, and reports quarterly workforce metrics to PERC.

Training delivery partners: NAPTIN (National Power Training Institute), polytechnics (federal and state), technical colleges and TVET institutions, engineering faculties at universities, DisCo and GenCo in-house academies, Meter Asset Providers and equipment OEMs (mandatory training components), PTI (Port Harcourt Training Institute) for gas infrastructure roles.

## 11.2 Training Tracks and Targets

| Role | 5-Year Target | Primary Training Institution | Certification Body | Phase |
|---|---|---|---|---|
| Meter installers | 15,000–20,000 | TVET, polytechnics, MAP vendors, NAPTIN | NESCB | 0–1 (Yrs 1–2) |
| Distribution linemen | 8,000–12,000 | NAPTIN, polytechnics, DisCo in-house | NEMSA (EA S.82) | 1–4 (Yrs 1–3) |
| Substation / SCADA technicians | 2,000–3,000 | NAPTIN, TCN facilities, OEM programmes | NEMSA + OEM | 0–2 (Yrs 1–2) |

| | | | | |
|---|---|---|---|---|
| Distribution engineers | 3,000–5,000 | Engineering faculties, NAPTIN | COREN + NESCB | 1–4 (Yrs 2–4) |
| Data platform operators | 1,500–2,500 | Universities, private tech training, DFI-funded | NEDO internal + NESCB | 0–4 (Yrs 1–3) |
| Utility data analysts | 800–1,200 | Universities, NAPTIN analytics programme | NESCB | 1–3 (Yrs 2–4) |
| Feeder monitoring teams | 3,000–5,000 | NERC, NEMSA, third-party audit firms | NERC/NEMSA | 0–1 (Yrs 1–2) |
| Transformer / substation techs | 2,000–3,500 | NAPTIN, DisCo academies, OEM training | NEMSA + NESCB | 1–3 (Yrs 1–3) |
| Ombudsman case officers | 300–500 | Law schools, NERC regulatory training | NEO internal | 1–2 (Yrs 1–2) |
| Forensic utility auditors | 200–400 | Accountancy bodies, NEMSA, EFCC training | NESCB + ICAN | 1–3 (Yrs 2–4) |
| Enforcement support specialists | 300–500 | NPS, NSCDC, NERC training | NERC + NSCDC | 1–3 (Yrs 1–3) |
| Gas infrastructure technicians | 5,000–8,000 | PTI, NNPC Gas, IOC programmes | DPR/NUPRC | 2–4 (Yrs 2–5) |
| Security personnel | 4,000–6,000 | Police, military, private security, community | Security agencies | 1–3 (Yrs 1–3) |

## 11.3 Curriculum and Syllabus Reform

Existing syllabi at NAPTIN and most polytechnics are not aligned to current sector needs. The NESCB will commission revised curricula for each role category above, developed in partnership with DisCos, GenCos, TCN, and international technical partners. Revised syllabi will cover: modern metering systems and AMI architectures; SCADA/EMS operation and diagnostics; utility data management and anomaly detection; regulatory and compliance frameworks under the Electricity Act 2023; customer service and dispute resolution; anti-corruption ethics for sector roles.

Curriculum validation. Each revised curriculum is reviewed by NESCB, at least one international utility body (such as the International Electrotechnical Commission or the Energy Institute UK), and NEMSA before accreditation. Review cycle: every 3 years.

## 11.4 Accreditation and Certification Pathways

Two-tier certification. Tier 1 — Trade certification: for practical installation and field roles (meter installers, linemen, substation technicians). Assessed by practical examination and

supervised deployment hours. Issued by NESCB. Required for employment on any government or DFI-funded electricity contract. Tier 2 — Professional certification: for engineers, data analysts, forensic auditors, and senior roles. Requires a Tier 1 foundation plus additional academic and professional assessment. Co-issued by NESCB and the relevant professional body (COREN, ICAN, NERC).

National skills register. NESCB maintains a digitally searchable register of all certified individuals in the sector. Employers and contracting authorities can verify certification status. Certification is revocable for conviction of electricity sector offences.

Portability. Certifications are nationally recognised and portable across DisCos, GenCos, TCN, state utilities, and private operators. This reduces the incentive for DisCos to hoard trained staff and increases labour market mobility.

## 11.5 Contractor Training Obligations

All major contracts above $10 million must include mandatory workforce training components. The minimum standard: at least 80% of installation, maintenance, and operational workforce must be Nigerian nationals. At least 40% of Nigerian nationals employed on the contract must receive NESCB-accredited training as part of the contract scope.

DisCo apprenticeship requirement. Each DisCo is required to maintain an apprenticeship programme of at least 500 apprentices (scaled to customer base) as a licence compliance condition, assessed annually by NEMSA.

OEM technology transfer. All equipment procurement contracts above $5 million must include a technology transfer component: OEM partners must deliver training programmes for Nigerian technicians on the procured equipment as a deliverable within the contract. This is enforceable as a contract condition and monitored by NEDO.

## 11.6 Reducing Dependence on Foreign Technical Execution

Nigeria's current dependence on foreign technical consultants for SCADA operation, smart metering deployment, and data platform management is a cost risk, a sovereignty risk, and a bottleneck risk. The NEWP targets 80% Nigerian workforce participation across all major technical roles by Year 4. This is achievable because the roles require technical competence, not rare specialisms — the global supply chain for all technologies involved is well-established, and the training pathways are mature.

Year 1: Foreign technical leads permitted for SCADA completion and NEDP build. Nigerian counterpart trainees required on every foreign-led team. Year 3: Nigerian engineers must hold at least co-lead status on all SCADA and NEDP operational roles. Year 5: Foreign

technical staff permitted only for specialised OEM roles that have no Nigerian equivalent certification pathway. All other operational and supervisory roles must be Nigerian national.

# 12. Consumer Protection and Trust-Building

Consumer protection is the institutional face of the fairness doctrine. If the plan starts with 'fairness first,' then consumer protection is how fairness is made real, visible, and enforceable. A citizen who has a meter, receives an accurate bill, has a working complaint channel, and can see their DisCo's performance statistics has no legitimate reason to bypass the formal system. That is the condition this section creates.

## 12.1 National Electricity Ombudsman

| | |
|---|---|
| **Independence** | Appointed by the President on recommendation of the National Assembly. Fixed 5-year term. Funded by market participant levy, not NERC or Ministry budgets. Cannot be removed except by National Assembly resolution. |
| **Powers** | Investigate complaints. Order refunds. Impose financial penalties on DisCos for billing abuse. Publish quarterly performance reports. Refer criminal matters to the Electricity Division. Issue systemic finding orders requiring DisCo-wide process changes. |
| **Complaint timelines** | DisCo acknowledgement: 72 hours. Billing dispute resolution: 7 working days. Escalation to NEO: 14 days. NEO determination: 30 days. |
| **Public reporting** | Quarterly: complaints received by DisCo, resolution rates, penalty amounts, systemic issues, refunds ordered. Published on the NEDP public dashboard. |
| **Non-compliance** | DisCos that fail to respond to NEO orders within the specified timeline face automatic financial penalty at the next PMF assessment. Repeat failure is a Tier 3 KPI trigger under Annex E. |

## 12.2 Community Service Compacts

Contractual instruments between DisCos and organised community structures: landlords' associations, market unions, estate committees, traditional authorities, local government units. The DisCo commits to minimum service hours, metering, accurate billing, and maintenance response SLAs. The community commits to payment discipline and theft reduction targets within the feeder zone.

Where losses drop and payment rises against the compact's baseline, the community receives visible benefits under a pre-agreed schedule: transformer upgrade, extended service hours, street lighting, or a credit toward community meter installation. The compact is legally binding. DisCo non-performance triggers NEO complaint rights at the community level. Community non-performance allows DisCo to suspend compact benefits.

Compact design is standardised by NERC (proposed model compact) but tailored by zone. NEDO provides feeder-level baseline data to anchor the compact's performance targets in verified numbers.

## 12.3 Billing Transition Protocol

| | |
|---|---|
| **Grace period** | First 3 billing cycles after meter installation: no disconnection for payment disputes arising from the transition. |
| **Reconciliation** | Where metered consumption is significantly lower than prior estimated billing, the customer is entitled to a credit equal to overpayment in the preceding 6 months. |
| **Where consumption is higher** | Customer transitions to metered rate without back-billing for the pre-metered period. |
| **Communication** | Written explanation in English and relevant local language of tariff band, meter reading process, complaint channel, and billing transition protocol. |
| **Dispute escalation** | Any transition dispute unresolved within 7 days escalates to the NEO automatically if the customer requests it. No barrier to escalation. |

# 13. Subsidy Transition and End State

Nigeria must define the endpoint. Without a terminal condition, electricity subsidy becomes permanent, which is exactly what happened with fuel subsidy. The political incentive to freeze tariffs never disappears; it must be structurally constrained.

## 13.1 Preconditions for Major Subsidy Withdrawal

| | |
|---|---|
| **Metering coverage** | 90%+ of active customers metered (currently ~54%). |
| **Feeder metering** | 100% of feeders with real-time energy accounting. |
| **MDA compliance** | All federal and state MDAs on prepaid or auto-deduct. |
| **Collection efficiency** | DisCo collection exceeds 85% (currently ~74–76%). |
| **Service quality** | Band A average exceeds 20 hours/day. Service bands independently audited. |
| **Complaint resolution** | NEO operational. Customer complaint resolution rate above 80% within 7 days. |

## 13.2 Transition Sequence

| | |
|---|---|
| **Years 1–2** | Aggressive metering. MDAs prepaid. High-income and commercial users moved toward cost-reflective pricing. Hidden subsidy published as budget line. Lifeline tariff designed. PCAF operationalised (EA 2023, S.159). |
| **Year 3** | Major subsidy rollback for metered users with measurable service. Targeted subsidy retained for verified lifeline users only. |
| **Years 4–5** | Residual subsidy narrowed. Market pricing for most users. Direct welfare replaces electricity price distortion. |

## 13.3 Terminal Condition

Subsidy reaches zero for all non-lifeline users when all preconditions are met and have been sustained for at least two consecutive quarters, verified by independent audit. Lifeline tariff becomes the only remaining public support, funded through a transparent budget line with published beneficiary data. The PCAF (EA 2023, Section 159) is the statutory vehicle for lifeline support. Nigeria must name this endpoint in legislation; without a named endpoint, the subsidy has no natural termination mechanism.

# 14. Irreversibility Architecture

One weakness in previous Nigerian electricity reforms is that they were reversible. A new minister, a new government, a political calculation about tariffs, or quiet non-enforcement of a regulatory order could unwind years of progress within months. This plan is designed with specific lock-in mechanisms that make reversal structurally costly, not merely politically inconvenient.

## 14.1 How MDA Prepaid Stops Arrears from Returning

Once federal MDAs are on prepaid metering with automatic budget deduction, the mechanism for accumulating electricity debt no longer exists. The deduction happens at source, through the OAGF, before funds reach the MDA's operational budget. Reversing this requires an affirmative political decision to remove the prepaid mandate, which is politically costly once the system is operational and the cost savings are documented. The MDA payment dashboard ensures that any attempt to quietly reverse prepaid status is immediately visible in the public data.

## 14.2 How Public Dashboards Change the Political Cost of Opacity

Once feeder-level performance data is publicly available, the political cost of a minister or governor attempting to hide DisCo underperformance increases substantially. Journalists, civil society organisations, academic researchers, and opposition politicians can all access the same underlying data. A government that shuts down the public dashboard or allows data submission to collapse faces the same political cost as a government that openly suppresses an independent audit. The NEDP is designed so that non-submission by DisCos is automatically visible as a compliance failure, not as an invisible event.

## 14.3 How Recapitalisation Powers Change Owner Behaviour

The EA Amendment Bill's compulsory recapitalisation mechanism creates a credible threat for the first time. DisCo owners who have previously treated their licences as low-risk assets generating rent from an opaque, tariff-distorted market now face the possibility of share dilution, managed receivership, or forced re-privatisation. The mere existence of this power changes behaviour before it is exercised. The DisCo Transition Protocol's seven quantitative KPIs give owners a clear, observable target. The 12-month cure period for Tier 2 DisCos is long enough to be fair; the automatic Tier 3 trigger for failure to move a single KPI is short enough to be serious.

## 14.4 How the PSRV Creates Creditor Pressure

Once the PSRV issues government-backed notes to clear GenCo receivables, those notes create a new class of creditors with a direct claim against the electricity sector's revenue stream. Bond holders have an institutional incentive to monitor sector performance and

resist policy reversals that would impair recovery. The PSRV's 7-year sunset clause creates a defined accountability horizon. Ratings assigned to PSRV bonds by international credit agencies function as an ongoing market signal of sector health.

## 14.5 How Data Visibility, Fairness, and Complaint Systems Create a New Equilibrium

The reform creates a new population of stakeholders with an interest in maintaining it: metered customers who now receive accurate bills and have a functioning complaint mechanism; communities that have signed service compacts with measurable commitments on both sides; industrial users who have left the grid and built captive power but remain nominally connected to the formal system through eligible customer arrangements; state governments that have assumed regulatory authority and built institutions around it.

Each of these groups has something to lose from reversal. They are not large individual actors, but in aggregate they represent a diffuse pressure against regression that did not exist before the reform began. This is the political economy of irreversibility: reform creates its own constituency.

## 15. Climate and Energy Transition

This section is financial and strategic, not ideological. Nigeria's ability to attract the $37–49 billion required depends significantly on climate finance flows. A plan that adds gas-fired generation without addressing the transition pathway will find major financing channels closed.

| | |
|---|---|
| **Gas (short/medium term)** | Gas remains the backbone of thermal generation for 10–15 years. New gas investments must be designed for flexibility (combined-cycle with hydrogen blending potential). |
| **Renewables** | National Energy Compact targets 50% renewable share by 2030. Off-grid programme ($3–5bn) is predominantly renewable. EA 2023, Part XI, Sections 161–175. |
| **Battery storage** | Relevant for specific applications: solar intermittency, critical load backup, frequency regulation. Not yet a national strategy priority. |
| **Climate finance** | Green Climate Fund, bilateral climate finance, carbon credits. Compatible with gas-heavy near-term if framed as transition strategy with measurable targets. |

## 16. Regional and Strategic Context

| | |
|---|---|
| **West African Power Pool** | Nigeria is a net electricity exporter to Niger, Benin, Togo. Reform must not disrupt export capacity. Regional interconnection provides market and physical diversification. |
| **Trans-Saharan Gas Pipeline** | Proposed pipeline (Nigeria-Niger-Algeria) has implications for domestic gas availability. PIA 2021 domestic gas obligations must be enforced. |
| **DFI competition** | Nigeria competes with Kenya, South Africa, Morocco, Egypt for limited DFI capital. Demonstrable reform credibility reduces the cost of capital. |
| **Strategic positioning** | Achieving 60% of this plan's targets would give Nigeria the largest, most data-transparent electricity system in West Africa. |

## 17. Risk Analysis

| Risk | Likelihood | Impact | Mitigation | Lead |
|---|---|---|---|---|
| Political interference in tariff reform | High | Critical | Trigger-based withdrawal, transparent budget subsidy, PCAF, public dashboards showing fiscal cost of delay | NERC / MoF |
| DisCo resistance to reform | High | High | EA Amendment Bill with compulsory recapitalisation; PMF evidence base; DisCo Transition Protocol (Annex E); Resistance Map (Section 9) | NERC / PERC |
| Metering deployment delay | Medium-High | High | Multiple frameworks (MAP, NMMP, MAF), workforce programme, PERC weekly monitoring | NEDO / DisCos |
| Gas supply disruption | High | Critical | Gas Payment Assurance Facility, pipeline security (Annex F), PIA 2021 enforcement, generation diversification | FMP / NNPC |
| Federal-state tariff conflict | Medium | High | Federal-State Tariff Interface Rules, Forum of Regulators, NEDP transparency | NERC / SERCs |
| Data platform integrity | Medium | Critical | Mandatory submission as licence condition, immutable audit log, anomaly detection, criminal penalties for falsification | NEDO / NERC |
| Financing shortfall | Medium-High | High | Phased investment, bankability classification, performance guarantees, carbon credit monetisation | MoF / DFIs |
| Public resistance to subsidy removal | High | High | Fairness doctrine: metering first, complaint resolution second, service improvement third, then price truth. PCAF for vulnerable users | NERC / MoF |
| Institutional turf wars | Medium | Medium | Clear mandate definitions, PERC as coordination (not regulatory) body, NISO-NEDO integration protocol | PERC |
| Security deterioration | High | Critical | Power Infrastructure Security Framework (Annex F), Electricity Division for rapid | Security / PERC |

| | | | | |
|---|---|---|---|---|
| | | | prosecution, economic sabotage classification | |
| Internal collusion undermining data | High | High | Immutable audit log, tamper alerts bypassing DisCo management, individual criminal liability for falsification, whistle-blower protection | NEDO / NERC |
| Workforce capacity gap | Medium | High | NEWP five-year programme with 37,000–55,000 new certified workers; contractor training obligations; OEM technology transfer | NEDO / NESCB |

## 18. Conclusion

Nigeria's electricity problem is not solved by one large intervention. It is solved by putting the system under control first, then fixing it. Most reforms attempt to fix before control. That is why they fail.

| | |
|---|---|
| 01 | Make the system visible. |
| 02 | Increase revenue immediately. |
| 03 | Clear debt with control in place. |
| 04 | Enforce after fairness is established. |
| 05 | Build infrastructure where it can be delivered and paid for. |
| 06 | Move to targeted subsidies and then price truth. |
| 07 | Decentralise aggressively through state markets and off-grid power. |
| 08 | Let competition discipline the market. |

To accelerate power reform in Nigeria, five things are required simultaneously: deterrence, fairness, speed, visibility, and local deal-making. Most reform efforts bring one or two. Deterrence without fairness becomes oppression. Fairness without speed becomes bureaucracy. Speed without visibility becomes propaganda. Visibility without local deal-making becomes empty dashboards. Local deal-making without deterrence becomes unserious.

**The system must be fair enough to defend, hard enough to circumvent, local enough to feel real, and fast enough to change behaviour before cynicism returns.**

*Nigeria does not need more electricity plans. It needs a system people cannot circumvent, cannot ignore, and can finally trust.*

# 19. Methodology and Sources Note

## Authorship

This document was prepared under the editorial direction of the Office of the Citizen (officeofthecitizen.org). The Office of the Citizen is an evidence-based accountability and civic documentation initiative focused on Nigeria. The organisation operate independently of government, party, and commercial electricity sector interests.

## Data Sources

This document draws on the following primary sources: NERC Quarterly Reports (Q1–Q4 2024, Q1–Q2 2025), NERC Annual Report 2024, NERC Performance Monitoring Framework Order (July 2024) and Addendum-1 (December 2024), World Bank project documentation (PSRO, DISREP, DARES, Nigeria Transmission Programme), Federal Ministry of Power publications, Nigeria National Energy Compact, AfDB Regulatory Governance Index 2024, UNDP Nigeria sector analysis, and industry reporting.

## Date Range

Performance data covers the period 2024–2025. Legal references are to legislation as at March 2026. Cost estimates use prevailing exchange rates and pricing as at Q1 2026.

## Estimates and Assumptions

Certain figures in this document are author estimates rather than sourced data. These include: per-meter installation cost (~$250), total programme cost ranges ($37–49 billion), workforce numbers, and some financing split percentages. All estimates are clearly marked. Cost ranges are indicative and should be validated through detailed feasibility studies before procurement.

## Current Law vs Proposed Reform

The document distinguishes between reforms achievable under existing law (EA 2023, PIA 2021, Constitution as amended, NERC regulations) and those requiring new legislation or regulatory instruments. Proposed new instruments are clearly labelled throughout the document and summarised in Section 5.2.

## Disclaimer

This document does not constitute legal advice. Readers who intend to rely on legal references for decision-making should seek independent legal opinion. All legislative references are to Nigerian statutes as at March 2026.

# Annex A: Implementation Matrix

*Technical Annex. This matrix converts each major action into an accountable assignment.*

| Action | Lead | Legal Basis | Funding | Timeline | KPI | Escalation |
|---|---|---|---|---|---|---|
| Establish PERC | Presidency | Constitution S.5 | Public budget | Month 1 | Council constituted, first meeting | Presidential intervention |
| Establish NEDO | PERC | PERC instrument | Public + DFI | Month 2 | Office operational | PERC to Presidency |
| Launch NEDP | NEDO | Proposed Data Regs | DFI ($200–400m) | Months 3–6 | All DisCos submitting data | Licence review (EA S.76) |
| Top 200 feeder metering | NEDO + DisCos | EA S.105; PMF Order | DFI + vendor | Months 1–6 | 200 feeders real-time | PMF penalty (5% admin) |
| MDA prepaid metering | NEDO + OAGF | Executive directive | Public budget | Months 1–6 | All MDAs prepaid | OAGF enforcement |
| Energy flow audit | NEDO + NERC | EA S.56(3) | DFI + public | Months 3–6 | Published audit report | PERC escalation |
| Mass meter rollout | DisCos + MAPs | EA S.105; MAP/NMMP/MAF | Vendor + MAF | Months 6–18 | 2m meters; >65% rate | NERC; PERC weekly |
| Amnesty window | NERC | Proposed Order (EA S.96) | Cost-neutral | 60–90 days | Connections regularised | Post-amnesty enforcement |
| Establish NEO | Nat'l Assembly | Proposed NEO Act | Market levy | Months 6–12 | NEO operational | Nat'l Assembly oversight |
| Create PSRV | MoF / Presidency | Proposed PSRV Act | Sovereign bonds | Months 6–12 | First arrears tranche settled | PERC; Nat'l Assembly |
| Gas Payment Facility | NBET / FMP | EA S.7; PIA S.104 | Tariff + guarantee | Months 6–18 | On-time gas payment >90% | PIA enforcement |

| | | | | | | |
|---|---|---|---|---|---|---|
| Electricity Division | CJ FHC / AG | FHC Act S.44; EA S.227 | Judiciary budget | Months 6–12 | Division operational; first cases | AG escalation |
| DisCo assessment | NERC | EA S.73–95; PMF | NERC budget | Months 6–12 | All 11 DisCos classified | Tier 2/3 triggers (Annex E) |
| NEWP launch | NEDO / NESCB | EA S.82; NEMSA | DFI + sovereign | Months 3–12 | Training contracts signed, first cohort enrolled | PERC escalation |
| Distribution rehab | DisCos + PERC | EA Part VI | DFI + blended | Months 12–48 | ATC&C <25% on priority feeders | DisCo Protocol (Annex E) |
| SCADA completion | TCN / NISO | EA S.26–42 | WB + sovereign | Months 6–24 | SCADA 100% operational | PERC; TCN accountability |
| New generation | Private / IPPs | EA Part IV | Commercial | Months 12–48 | Commissioned with signed contracts | Market forces; NERC |
| Subsidy withdrawal | NERC / MoF | EA S.136–137, S.159 | Budget savings | Months 24–48 | Cost-reflective for non-lifeline | Preconditions gate; independent audit |

## Annex B: Financing Breakdown

*Technical Annex. Detailed financing splits by source category.*

| Component | Amount | Sovereign | Concessional | Blended | Commercial | Preconditions |
|---|---|---|---|---|---|---|
| Legacy debt | ₦2.8tn | Primary (bonds) | Partial | — | — | PSRV legislation; verified audit |
| Working capital | $2–3bn | Guarantee | Primary (WB/AfDB) | — | — | Payment waterfall design |
| Metering | $1.4–1.8bn | Credit guarantees | WB (DARES/DISREP) | MAP vendor | Vendor contracts | MAP/NMMP/MAF frameworks |
| Distribution | $10–12bn | Co-fund (20%) | AfDB/IFC (30%) | Primary (40%) | After tariff (10%) | Tariff reform; DisCo performance |
| Transmission | $6–8bn | Primary (50%) | WB (40%) | PPP (10%) | — | TCN institutional reform |
| Gas infra | $5–7bn | Co-fund (20%) | DFI (20%) | Primary (30%) | IOC equity (30%) | PIA domestic gas enforcement |
| Generation | $9–12bn | — | Risk cover | Partial (20%) | Primary (80%) | Signed gas/offtake/evacuation |
| Off-grid | $3–5bn | Subsidies (20%) | DARES/DFI (40%) | Primary (30%) | IPPs (10%) | State electricity laws |
| Digitisation | $2–4bn | Co-fund (30%) | Primary (50%) | Tech (20%) | — | NEDP design approved |
| Workforce | $0.2–0.4bn | Primary (60%) | Grant (40%) | — | — | Training contracts |

# Annex C: Data and Monitoring Architecture

*Technical Annex. Full architecture specification in Section 10. This table summarises governance parameters.*

| | |
|---|---|
| **Owner** | NEDO (build phase, 0–5 years). Transitions to NISO or independent Data Authority. |
| **Architecture** | Central federal platform with state integration nodes. Three layers: (1) generation/transmission (GenCos, TCN, NISO); (2) distribution/retail (DisCos, SERCs); (3) customer/metering (DisCos, MAPs, vendors). |
| **Data submission** | Mandatory for all licenced participants. Hourly for priority feeder and real-time grid data. Daily for all generation/transmission. Monthly for retail/billing. Non-submission triggers licence review (EA S.76). |
| **Interoperability** | Open API standards. Prescribed data formats. No proprietary lock-in. Schema published and version-controlled. |
| **Audit log** | Immutable append-only ledger. Original data preserved regardless of amendments. Chain-of-custody interface for Electricity Division. |
| **Anomaly detection** | Automated weekly scan for loss spikes, billing-to-collection divergence, fabrication patterns. Flags routed to NERC within 48 hours. |
| **Public dashboards** | Feeder-level: energy received, billed, collected, hours supplied, loss %. MDA payment dashboard. Enforcement action register. NEO case dashboard. |
| **Independent audit** | Third-party firm, rotated every 3 years. Quarterly integrity audits. Penetration test annually. Results published. |
| **Cybersecurity** | Critical national infrastructure classification. End-to-end encryption, MFA, network segmentation, IDS/IPS. Incident disclosure within 72 hours. |
| **Connectivity fallback** | Hierarchy: fibre → 4G/LTE → VSAT. Store-and-forward mesh for low-connectivity zones. Manual submission protocol for outages beyond 72 hours. |
| **Data falsification** | Proposed criminal offence equivalent to financial fraud. Immutable log makes post-hoc falsification detectable. |

# Annex D: Workforce Mobilisation Plan

*Technical Annex. Full programme design in Section 11. This table summarises deployment parameters.*

| Category | 5-Yr Target | Training Pipeline | Certification | Phase |
|---|---|---|---|---|
| Meter installers | 15,000–20,000 | TVET, polytechnics, MAP vendors, NAPTIN | NESCB | 0–1 (Yrs 1–2) |
| Distribution linemen | 8,000–12,000 | NAPTIN, polytechnics, DisCo in-house | NEMSA (EAS.82) | 1–4 (Yrs 1–3) |
| Substation / SCADA techs | 2,000–3,000 | NAPTIN, TCN facilities, OEM programmes | NEMSA + OEM | 0–2 (Yrs 1–2) |
| Distribution engineers | 3,000–5,000 | Engineering faculties, NAPTIN | COREN + NESCB | 1–4 (Yrs 2–4) |
| Data platform operators | 1,500–2,500 | Universities, private tech training, DFI | NEDO + NESCB | 0–4 (Yrs 1–3) |
| Utility data analysts | 800–1,200 | Universities, NAPTIN analytics | NESCB | 1–3 (Yrs 2–4) |
| Feeder monitoring teams | 3,000–5,000 | NERC, NEMSA, third-party audit firms | NERC/NEMSA | 0–1 (Yrs 1–2) |
| Ombudsman case officers | 300–500 | Law schools, NERC regulatory training | NEO internal | 1–2 (Yrs 1–2) |
| Forensic utility auditors | 200–400 | Accountancy bodies, NEMSA, EFCC | NESCB + ICAN | 1–3 (Yrs 2–4) |
| Enforcement support | 300–500 | NPS, NSCDC, NERC training | NERC + NSCDC | 1–3 (Yrs 1–3) |
| Gas infrastructure | 5,000–8,000 | PTI, NNPC Gas, IOC programmes | DPR/NUPRC | 2–4 (Yrs 2–5) |
| Security personnel | 4,000–6,000 | Police, military, private security | Security agencies | 1–3 (Yrs 1–3) |

All major contracts above $10 million must include mandatory workforce training components. At least 80% of the installation and maintenance workforce must be Nigerian nationals. Each DisCo is required to maintain an apprenticeship programme (minimum 500 apprentices, scaled to customer base) as a licence compliance condition.

# Annex E: DisCo Transition Protocol

*Technical Annex. Formal performance review, classification, and ownership transition mechanism.*

## E.1 Quantitative Thresholds

| KPI | Tier 1 (Performing) | Tier 2 (Underperforming) | Tier 3 (Failing) |
|---|---|---|---|
| ATC&C losses | <25% | 25–35% | >35% |
| Collection efficiency | >85% | 70–85% | <70% |
| Metering coverage | >80% | 60–80% | <60% |
| Average service hours (Band A) | >20 hrs/day | 16–20 hrs/day | <16 hrs/day |
| Infrastructure investment (% revenue) | >15% | 8–15% | <8% |
| Complaint resolution (7 days) | >80% | 60–80% | <60% |
| NBET remittance compliance | >95% | 85–95% | <85% |

*Classification based on worst-performing two KPIs. A DisCo scoring Tier 1 on six KPIs but Tier 3 on one is classified Tier 3.*

## E.2 Classification and Response

| | |
|---|---|
| **Tier 1: Performing** | Priority access to concessional finance, tariff flexibility, expanded licence territory, public recognition. |
| **Tier 2: Underperforming** | Formal Improvement Notice (EA S.74). 12-month remediation. Performance Improvement Plan within 30 days. Quarterly reviews. Failure to move at least one KPI to Tier 1 triggers Tier 3. |
| **Tier 3: Failing** | Mandatory intervention: (1) Compulsory recapitalisation — inject capital in 90 days or face dilution; (2) Managed receivership — independent management appointed by NERC; (3) Re-privatisation through competitive tender. State buyout option available. |

## E.3 Treatment of Stakeholders

| | |
|---|---|
| **Shareholders** | Diluted if recapitalisation fails. Bought out at independently assessed fair value in re-privatisation. Right to challenge |

| | |
|---|---|
| | valuation through Electricity Division, but challenge does not stay the process. |
| **Lenders** | Loan agreements preserved. New management assumes debt. Lender consent required where change-of-control clauses exist. |
| **Employees** | Contracts preserved through transition. No mass terminations without Labour Act compliance. |

*Legal basis: Current: EA 2023, S.73–95, S.76. Proposed: EA Amendment Bill 2025 (compulsory recapitalisation, receivership). Precedent: BOFIA 2020, S.12; BPE under Privatisation Act.*

# Annex F: Power Infrastructure Security Framework

*Technical Annex.*

| | |
|---|---|
| **Gas-to-power corridors** | Federal protection. Military/police patrols (NSCDC pipeline mandate), community surveillance, drone monitoring, rapid-response teams. Gas pipeline vandalism classified as economic sabotage. |
| **Transmission** | Federal/TCN. Perimeter security at critical substations, intrusion detection, community alert networks, asset marking and tracking. |
| **Distribution** | State/DisCo. Anti-vandal transformer cages, sealed meter boxes, tamper-alert meters, community watch integration. |
| **ESOZs** | Dedicated security for designated industrial/commercial zones. Integrated with local police. 24-hour patrol. Funded by zone operator levies and public budget. |
| **Intelligence** | Joint intelligence cell (NSCDC, police, DSS). Regular threat assessments. Information sharing protocol between security agencies, TCN, DisCos, and gas companies. |
| **Court integration** | All infrastructure sabotage cases handled by Electricity Division. Evidence chain: incident report → forensic investigation → prosecution brief → court filing within 14 days. |

*Legal basis: EA 2023, Sections 217–227; NSCDC Act; Miscellaneous Offences Act; PIA 2021, Section 215.*

*This document was prepared under the editorial direction of the Office of the Citizen (officeofthecitizen.org). All legal references are to Nigerian legislation as at March 2026. Proposed new instruments are clearly marked. Data figures are sourced from NERC quarterly reports, NERC Annual Report 2024, World Bank documentation, and Federal Ministry of Power publications. This document does not constitute legal advice.*